

- 8 Egg cells are produced by many organisms.
The development of fish embryos can be studied using fertilised egg cells.
The photograph shows part of a beluga fish from the Caspian Sea.



(Source: © Design Pics / Alamy Stock Photo)

- (a) Explain how an egg cell is specialised for its function.

(2)

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(b) A student studied beluga fish cells undergoing mitosis and meiosis.

(i) Compare and contrast metaphase in mitosis and meiosis.

(3)

(ii) Some of the cells undergoing mitosis were from a beluga morula whereas some of the cells were from a beluga blastocyst.

Explain how the cells of the beluga morula change as they develop into the cells of the blastocyst.

(4)



- (c) Chickens produce eggs.

The effect of the age of a chicken on the mass of the eggs it laid was investigated.

A statistical test can be used to see if there is a significant correlation between the age of a chicken and the mass of her eggs.

The value generated by this statistical test will always be between -1.0 to $+1.0$.

The table shows the mass of eggs collected from chickens of different ages.

Age / weeks	Rank of age	Mass of egg / g	Rank of mass of egg	D	D^2
36	1	2.27	8	-7	49
39	2	1.91			
42	3	2.18	5.5	-2.5	6.25
45	4	2.28	9	-5	25
48	5	2.12	4	1	1
51	6	2.19	7	-1	1
54	7	2.18	5.5	1.5	2.25
57	8	2.09	3	5	25
60	9	2.03	2		

- (i) Calculate the value for $\sum D^2$.
Use the table to help you.

(2)

$$\sum D^2 = \dots\dots\dots$$



(ii) Calculate the correlation coefficient, r_s value, for this data using the formula:

(2)

$$r_s = 1 - \frac{6(\sum D^2)}{n(n^2 - 1)}$$

Answer

(iii) The null hypothesis is that there was no significant correlation between the age of the chicken and the mass of the eggs laid.

Another scientist replicated this study and gained a correlation coefficient value of 0.38.

The table shows the critical values at the 95% probability level.

Number of pairs of data	Critical values at the 0.05 % probability level
8	0.738
9	0.683
10	0.648

Explain whether the null hypothesis should be accepted or rejected.

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(Total for Question 8 = 15 marks)

TOTAL FOR PAPER = 80 MARKS

